

AD23421 / INTERNSHIP

REGULATIONS – 2023

B. TECH

II YEAR – IV SEMESTER

ACADEMIC YEAR: 2025 – 2026 (EVEN)



**DEPARTMENT OF ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE**

**Rajalakshmi Engineering College
Thandalam, Chennai-602105**



Rajalakshmi Engineering College Thandalam, Chennai.

Cover Letter

Registration Number : 2116241801290
Student Name : Tarika A G
Year / Semester : II / IV
Department : Artificial Intelligence and Data Science
Email ID : tarika.ag.2024.aids@rajalakshmi.edu.in
Phone Number : 8807807801

Date:

Subject: Application for Internship Opportunity in Your Organization

Respected Sir/Madam,

I am a student pursuing **B.Tech in Artificial Intelligence and Data Science** at **Rajalakshmi Engineering College, Chennai**, and I am writing to apply for an internship in your esteemed organization. I am eager to gain practical experience and apply my academic knowledge to real-world problems in software and technology development.

I am passionate about learning new technologies, working in a team environment, and contributing to the development of recent software and innovative solutions. I believe this internship will help me enhance my technical and analytical skills while supporting your organization's goals.

Thank you for your time and consideration.

Yours sincerely,



Rajalakshmi Engineering College Thandalam, Chennai.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Recommendation Letter for Internship

This is to certify that **Mr./Ms. Tarika A G**, a student of **B.Tech in Artificial Intelligence and Data Science, Semester IV, Year II**, at **Rajalakshmi Engineering College, Chennai**, for the **Academic Year 2025-2026**, is recommended for an ☐ **Online** / ☐ **Offline** internship program in your esteemed organization.

Tarika A G has been a responsible and disciplined student during their time at our institution. We believe they will benefit from and positively contribute to the internship program.

We strongly recommend **Tarika A G** for the **Online / Offline** internship program and wish them success in all future endeavors.

Sincerely,

Dr. J M Gnanasekar

Professor & Head,

Department of Artificial Intelligence and Data Science,

Rajalakshmi Engineering College (Autonomous),

Thandalam - 602 105, Chennai.

hod.aids@rajalakshmi.edu.in.

Date: _____



Rajalakshmi Engineering College Thandalam, Chennai.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

DATA SCIENCE

AD23421 – INTERNSHIP

Internship Report submitted by

REGISTRATION NUMBER : 241801290

STUDENT NAME : TARIKA A G

YEAR / SEMESTER : II / IV

INTERNSHIP ORGANIZATION / COMPANY : ALTRUISTRY INNOVATION PVT LTD

ONLINE / OFFLINE : ONLINE

TRAINER NAME : PAMAJULA PRAVALIKA

INTERNSHIP PERIOD : FROM : 16/12/2025 TO: 31/12/2025

INTERNSHIP DURATION : DAYS 15

(Approved / Not Approved)
FACULTY ADVISOR

(Approved / Not Approved)
EXAMINER 1

(Approved / Not Approved)
EXAMINER 2

(Approved / Not Approved)
HoD / AIDS

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1 Abstract / Executive Summary

The main objective of the internship was to develop a **Customer Churn Analytics Dashboard** to analyze telecom customer data and identify factors contributing to customer attrition.

The project involved data preprocessing, exploratory data analysis, and implementation of a machine learning classification model to predict customer churn. Key features such as tenure, contract type, monthly charges, and internet services were analyzed to understand churn patterns. The insights were presented through an interactive dashboard displaying important metrics such as churn rate, revenue loss, and segment-wise analysis.

The internship provided hands-on experience in Python programming, data visualization, and machine learning workflows. It also enhanced analytical thinking, problem-solving skills, and understanding of how data-driven solutions support business decisions, especially in customer retention strategies.

2 Introduction

2.1 About the company

Altruisty Innovation Pvt Ltd is a technology-driven organization that focuses on developing innovative solutions in emerging domains such as Artificial Intelligence, Data Science, Machine Learning, and software development. The company aims to bridge the gap between academic education and industry requirements by providing training programs, internships, and real-time project exposure to students. Through practical learning approaches and mentorship, the organization helps students gain industry-relevant technical skills and professional experience.

The organization works on projects involving intelligent data analysis, automation systems, predictive analytics, and web-based applications. It emphasizes hands-on learning through project-based training, allowing interns to understand real-world workflows followed in software and data science industries. During the internship period, structured guidance and support were provided to understand data science concepts and develop a practical machine learning project.

2.2 Purpose of the internship

The purpose of the internship was to gain practical exposure to the working environment of a technology-based organization and to understand how theoretical knowledge of data science and machine learning can be applied to solve real-world problems. The internship provided an opportunity to learn industry-relevant tools and technologies used in data analysis and machine learning development.

Another important purpose was to enhance programming knowledge, analytical thinking, and problem-solving skills through hands-on project implementation. By working on a Customer Churn Analytics Dashboard project, the internship helped in understanding how organizations use data-driven approaches to analyze customer attrition patterns, identify churn risk factors, and improve business decision-making through effective customer retention strategies.

3 Objectives of the Internship

The key objectives of the internship were:

- To understand the fundamentals of Data Science and Customer Analytics.
- To learn data collection, cleaning, and preprocessing techniques for real-world datasets.
- To perform exploratory data analysis (EDA) to identify churn patterns and customer behavior trends.
- To implement machine learning algorithms for customer churn prediction.
- To develop an interactive Customer Churn Analytics Dashboard for visualizing insights and key performance indicators.
- To gain practical experience in Python, data visualization tools, and predictive modeling workflows.

4 Brief Description of Work / Project

4.1 Project Title:

Customer Churn Analytics Dashboard

4.2 Project Description:

As part of the internship, a Customer Churn Analytics project was carried out using a telecom customer dataset to analyze customer attrition patterns and predict churn behavior. The project aimed to evaluate key factors influencing customer churn, such as tenure, contract type, monthly charges, internet services, and payment methods.

The workflow began with data cleaning and preprocessing, including handling missing values, converting data types, and feature preparation. Exploratory Data Analysis (EDA) was then performed to identify trends, correlations, and high-risk customer segments. Key performance indicators such as churn rate, revenue loss due to churn, and segment-wise churn distribution were calculated to support business insights.

A machine learning classification model was implemented to predict the likelihood of customer churn. The results were visualized through an interactive dashboard that displayed churn patterns, risk indicators, and actionable insights. The project combined data analysis, predictive modeling, and visualization techniques to provide a comprehensive data-driven solution for improving customer retention strategies.

4.3 Tools and Methods Used:

The project followed a structured data science workflow integrating data preprocessing, exploratory data analysis, machine learning modeling, and dashboard visualization. Python was used as the primary tool, along with libraries such as Pandas and NumPy for data handling, Matplotlib and Seaborn for visualization, and Scikit-learn for implementing machine learning algorithms.

Initially, the telecom customer dataset was imported and cleaned by handling missing values, correcting data types, and encoding categorical variables. Feature engineering techniques such as tenure grouping were applied to enhance analysis. Exploratory Data Analysis (EDA) was conducted to identify churn trends across contract types, monthly charges, internet services, and customer demographics.

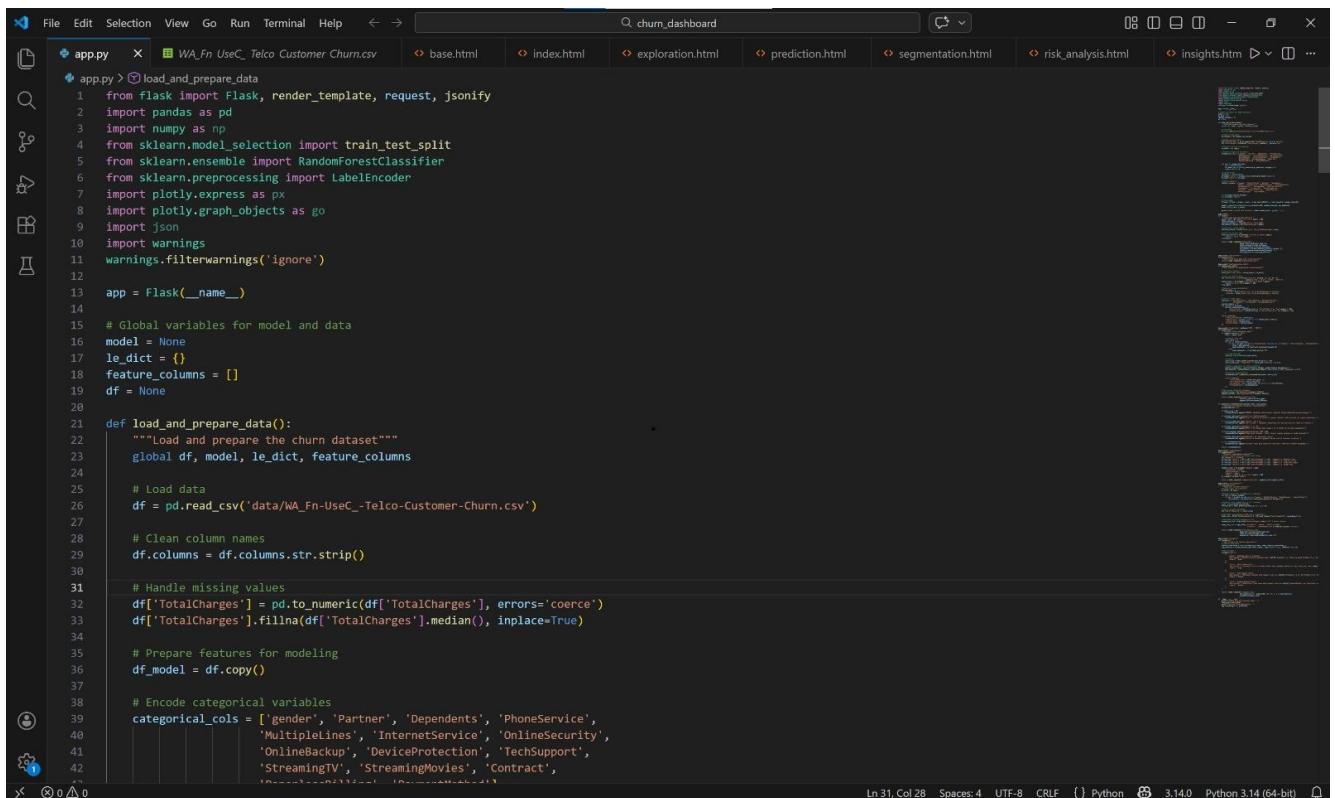
A classification model such as Logistic Regression Forest was implemented to predict customer churn. The dataset was split into training and testing sets, and model performance was evaluated using metrics like accuracy, precision, recall, and confusion matrix.

The final results and key insights were presented through an interactive Customer Churn Analytics Dashboard, highlighting churn rate, revenue loss, and high-risk customer segments. This integrated approach ensured a comprehensive and practical solution for understanding and reducing customer churn.

4.4 Outcomes / Results:

The project successfully developed a comprehensive Customer Churn Analytics Dashboard, highlighting patterns and trends in customer attrition across different segments. Visualizations were created to represent churn distribution based on contract type, tenure, monthly charges, and service categories. A detailed churn analysis was performed to identify high-risk customer groups and key factors contributing to customer loss.

The final dashboard quantified churn rate, revenue loss, and segment-wise churn variations, providing clear numerical insights into customer behavior. Performance metrics from the predictive model further supported the identification of customers likely to churn. The outputs included interactive dashboard reports, model evaluation results, and summarized business insights suitable for strategic decision-making and customer retention planning.



```
File Edit Selection View Go Run Terminal Help < -> churn_dashboard
app.py WA_Fn_UseC_Telco_Customer_Churn.csv base.html index.html exploration.html prediction.html segmentation.html risk_analysis.html insights.htm ...
app.py > load_and_prepare_data
1 from flask import Flask, render_template, request, jsonify
2 import pandas as pd
3 import numpy as np
4 from sklearn.model_selection import train_test_split
5 from sklearn.ensemble import RandomForestClassifier
6 from sklearn.preprocessing import LabelEncoder
7 import plotly.express as px
8 import plotly.graph_objects as go
9 import json
10 import warnings
11 warnings.filterwarnings('ignore')
12
13 app = Flask(__name__)
14
15 # Global variables for model and data
16 model = None
17 le_dict = {}
18 feature_columns = []
19 df = None
20
21 def load_and_prepare_data():
22     """Load and prepare the churn dataset"""
23     global df, model, le_dict, feature_columns
24
25     # Load data
26     df = pd.read_csv('data/WA_Fn-UseC-Telco-Customer-Churn.csv')
27
28     # Clean column names
29     df.columns = df.columns.str.strip()
30
31     # Handle missing values
32     df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
33     df['TotalCharges'].fillna(df['TotalCharges'].median(), inplace=True)
34
35     # Prepare features for modeling
36     df_model = df.copy()
37
38     # Encode categorical variables
39     categorical_cols = ['gender', 'Partner', 'Dependents', 'PhoneService',
40                        'MultipleLines', 'InternetService', 'OnlineSecurity',
41                        'OnlineBackup', 'DeviceProtection', 'TechSupport',
42                        'StreamingTV', 'StreamingMovies', 'Contract',
43                        'PaymentMethod']
44     le_dict = {}
45     for col in categorical_cols:
46         le = LabelEncoder()
47         le.fit(df[col])
48         le_dict[col] = le
49         df_model[col] = le.transform(df[col])
50
51     # Train and save the model
52     X = df_model.drop('Churn', axis=1)
53     y = df_model['Churn']
54     X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
55     model = RandomForestClassifier()
56     model.fit(X_train, y_train)
57     model.save('churn_model.pkl')
```

Customer churn web application home interface with dashboard and navigation options for data exploration, prediction, segmentation, risk analysis, and insights.

```
File Edit Selection View Go Run Terminal Help ← → Q churn_dashboard ↻ 🔍 📄 📁 📂 📅 📆 📇 📈 📉 📊 📋 📌 📍 📎 📏 📐 📑 📒 📓 📔 📕 📖 📗 📙 📚 📛 📜 📝 📞 📟 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📽 📾 📿 📠 📡 📢 📣 📤 📥 📦 📧 📨 📩 📪 📫 📬 📭 📮 📯 📰 📱 📲 📳 📴 📵 📶 📷 📸 📹 📺 📻 📼 📿
app.py x WA_Fn UseC_Telco Customer Churn.csv base.html index.html exploration.html prediction.html segmentation.html risk_analysis.html insights.htm
app.py > load_and_prepare_data
139
140 @app.route('/prediction', methods=['GET', 'POST'])
141 def prediction():
142     """Customer churn prediction page"""
143     if request.method == 'POST':
144         data = request.json
145
146         # Prepare input data
147         input_data = {}
148         for col in feature_columns:
149             if col in ie_dict and col != 'SeniorCitizen' and col not in ['tenure', 'MonthlyCharges', 'TotalCharges']:
150                 value = data.get(col, '')
151                 input_data[col] = ie_dict[col].transform([value])[0]
152             else:
153                 input_data[col] = float(data.get(col, 0))
154
155         # Create dataframe
156         input_df = pd.DataFrame(input_data)
157
158         # Predict
159         churn_prob = model.predict_proba(input_df)[0][1] * 100
160         churn_prediction = "High Risk" if churn_prob > 50 else "Low Risk"
161
162         # Feature importance for this prediction
163         feature_importance = dict(zip(feature_columns, model.feature_importances_))
164         top_features = sorted(feature_importance.items(), key=lambda x: x[1], reverse=True)[:5]
165
166         # Generate recommendations
167         recommendations = generate_recommendations(data, churn_prob)
168
169         return jsonify({
170             'churn_probability': round(churn_prob, 2),
171             'churn_prediction': churn_prediction,
172             'top_features': [(k, round(v*100, 2)) for k, v in top_features],
173             'recommendations': recommendations
174         })
175
176         # Get unique values for dropdowns
177         contract_types = df['Contract'].unique().tolist()
178         payment_methods = df['PaymentMethod'].unique().tolist()
179
```

Customer churn prediction interface displaying the churn probability, risk level, and recommended retention strategies based on customer input.

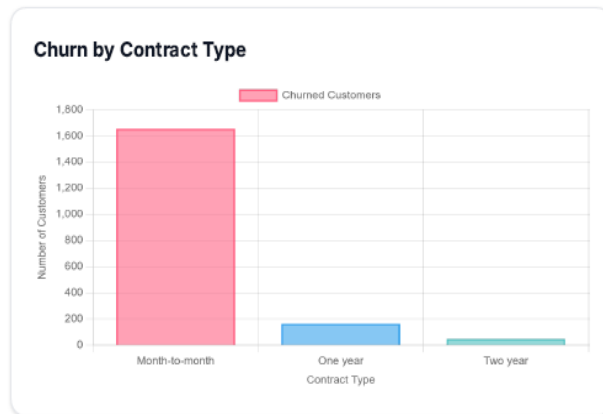
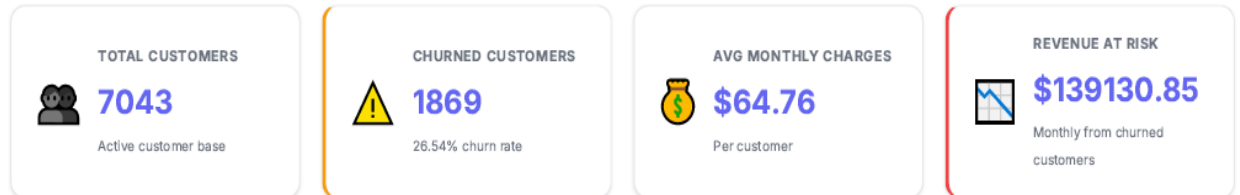
```
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app.py x WA_Fn UseC_Telco Customer Churn.csv base.html index.html exploration.html prediction.html segmentation.html risk_analysis.html insights.htm
app.py > load_and_prepare_data
141 def prediction():
175
176     # Get unique values for dropdowns
177     contract_types = df['Contract'].unique().tolist()
178     payment_methods = df['PaymentMethod'].unique().tolist()
179
180     return render_template('prediction.html',
181                           contract_types=contract_types,
182                           payment_methods=payment_methods)
183
184 def generate_recommendations(customer_data, churn_prob):
185     """Generate personalized retention recommendations"""
186     recommendations = []
187
188     if churn_prob > 70:
189         recommendations.append("URGENT: Immediate intervention required. Assign dedicated account manager.")
190
191     if customer_data.get('Contract') == 'Month-to-month':
192         recommendations.append("Offer incentive to switch to annual contract (20% discount for 1-year commitment).")
193
194     if float(customer_data.get('tenure', 0)) < 12:
195         recommendations.append("New customer - implement onboarding call and satisfaction check at 6 months.")
196
197     if customer_data.get('TechSupport') == 'No':
198         recommendations.append("Offer free premium tech support for 3 months to increase engagement.")
199
200     if float(customer_data.get('MonthlyCharges', 0)) > 70:
201         recommendations.append("High-value customer - offer loyalty rewards program or bundle discount.")
202
203     if customer_data.get('PaymentMethod') == 'Electronic check':
204         recommendations.append("Switch to automatic payment method with 5% discount incentive.")
205
206     if not recommendations:
207         recommendations.append("Customer shows good retention indicators. Continue standard engagement.")
208
209     return recommendations
210
```

Customer churn prediction result interface showing churn probability, risk classification, and personalized recommendations for customer retention.

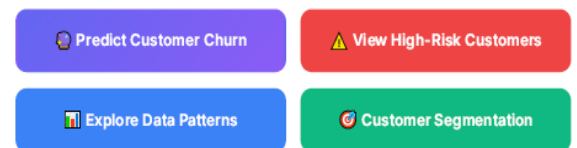


Customer Churn Dashboard

Overview of customer retention metrics and key performance indicators



Quick Actions



Key Insights



Customer churn web application dashboard interface displaying key metrics, visualizations, and navigation options for exploring different analysis pages.



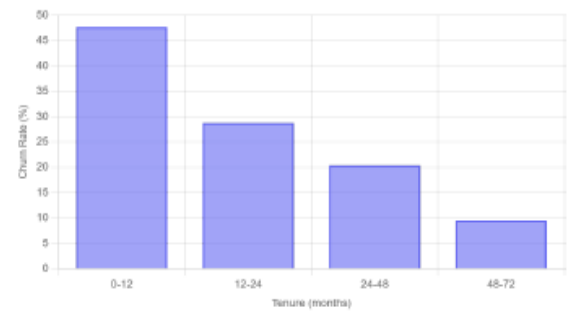
Data Exploration

Discover patterns and insights in customer behavior data

Churn Distribution



Churn by Tenure Group



Monthly Charges Distribution



Service Usage Impact on Churn



Customer churn data exploration interface showing charts and graphs used to analyze patterns and relationships in the telecom customer dataset.



Customer Churn Prediction

Enter customer details to predict churn probability and get retention recommendations

Customer Information

Gender	Senior Citizen
Male	No
Partner	Dependents
Yes	Yes
Tenure (months)	Phone Service
12	Yes
Multiple Lines	Internet Service
No	DSL
Online Security	Online Backup
No	No
Device Protection	Tech Support
No	No
Streaming TV	Streaming Movies
No	No
Contract	Paperless Billing
Month-to-month	Yes
Payment Method	Monthly Charges (\$)
Electronic check	65.00
Total Charges (\$)	
780.00	

Predict Churn Risk

Prediction Results



Low Risk

Top Contributing Factors

TotalCharges	15.79%
tenure	15.59%
MonthlyCharges	14.21%
Contract	13.25%
OnlineSecurity	7.86%

Retention Recommendations

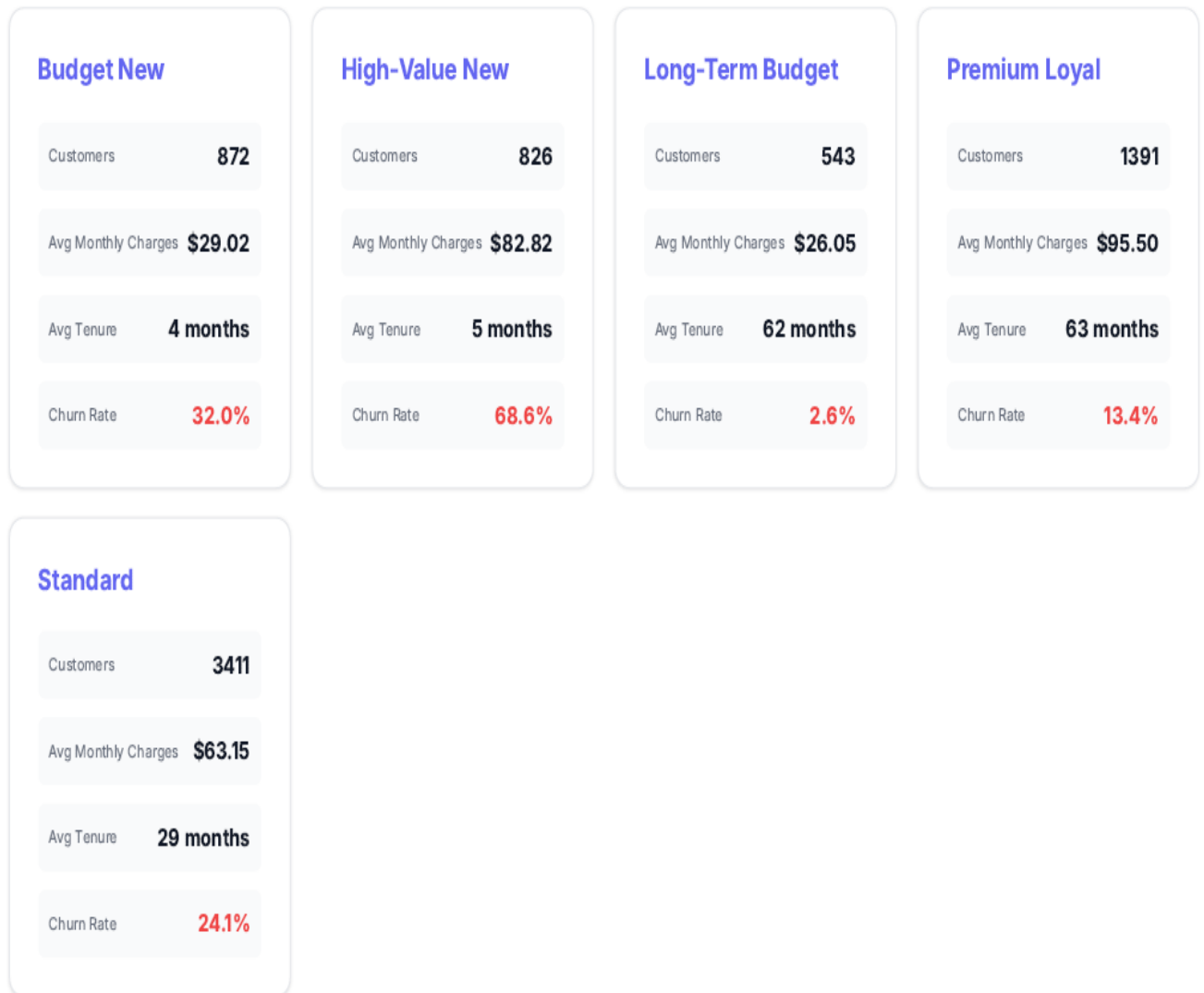
- Offer incentive to switch to annual contract (20% discount for 1-year commitment).
- Offer free premium tech support for 3 months to increase engagement.
- Switch to automatic payment method with 5% discount incentive.

Customer churn prediction interface allowing users to enter customer details and predict churn probability using the machine learning model.



Customer Segmentation

Understanding different customer groups and their characteristics



Customer segmentation interface categorizing customers into different groups based on their behavior and service usage patterns.

High-Risk Customer Analysis

Customers with high churn probability requiring immediate attention



HIGH-RISK CUSTOMERS

985

Churn probability > 60%



ANNUAL REVENUE AT RISK

\$915,013.80

Potential loss if no action taken

Top 50 High-Risk Customers

CUSTOMER ID	TENURE (MONTHS)	MONTHLY CHARGES	CONTRACT	CHURN PROBABILITY	ACTION
1069-XAIEM	1	\$85.05	Month-to-month	92.6%	Contact
9497-QCMMS	1	\$93.55	Month-to-month	92.4%	Contact
9300-AGZNL	1	\$94.00	Month-to-month	92.4%	Contact
1086-LXKFY	1	\$69.10	Month-to-month	92.2%	Contact
7216-EWTRS	1	\$100.80	Month-to-month	92.0%	Contact
4806-HIPDW	1	\$68.95	Month-to-month	91.8%	Contact
7908-QCBCA	1	\$69.40	Month-to-month	91.6%	Contact
9282-IZGQK	1	\$85.00	Month-to-month	91.6%	Contact
2506-TNFCO	1	\$69.55	Month-to-month	91.5%	Contact
3006-XIMLN	2	\$94.20	Month-to-month	91.3%	Contact
7660-HDPJV	1	\$69.20	Month-to-month	91.1%	Contact
9728-FTTVZ	1	\$69.20	Month-to-month	91.1%	Contact
0295-PPHDO	1	\$95.45	Month-to-month	91.1%	Contact
5299-SJCZT	3	\$101.55	Month-to-month	91.0%	Contact
5419-JPRRN	1	\$101.45	Month-to-month	91.0%	Contact
4501-EQDRN	1	\$69.70	Month-to-month	90.8%	Contact
1447-GIQMR	1	\$69.15	Month-to-month	90.7%	Contact
8603-IJWDN	1	\$86.60	Month-to-month	90.6%	Contact
2777-PHDEI	1	\$78.05	Month-to-month	90.6%	Contact
7665-TOALD	1	\$69.60	Month-to-month	90.6%	Contact
4910-GMJOT	1	\$94.60	Month-to-month	90.5%	Contact
9885-CSMWE	1	\$78.45	Month-to-month	90.4%	Contact
9019-QVLZD	1	\$79.55	Month-to-month	90.4%	Contact
6521-YYTYI	1	\$93.30	Month-to-month	90.3%	Contact
3158-MOERK	2	\$96.00	Month-to-month	90.3%	Contact

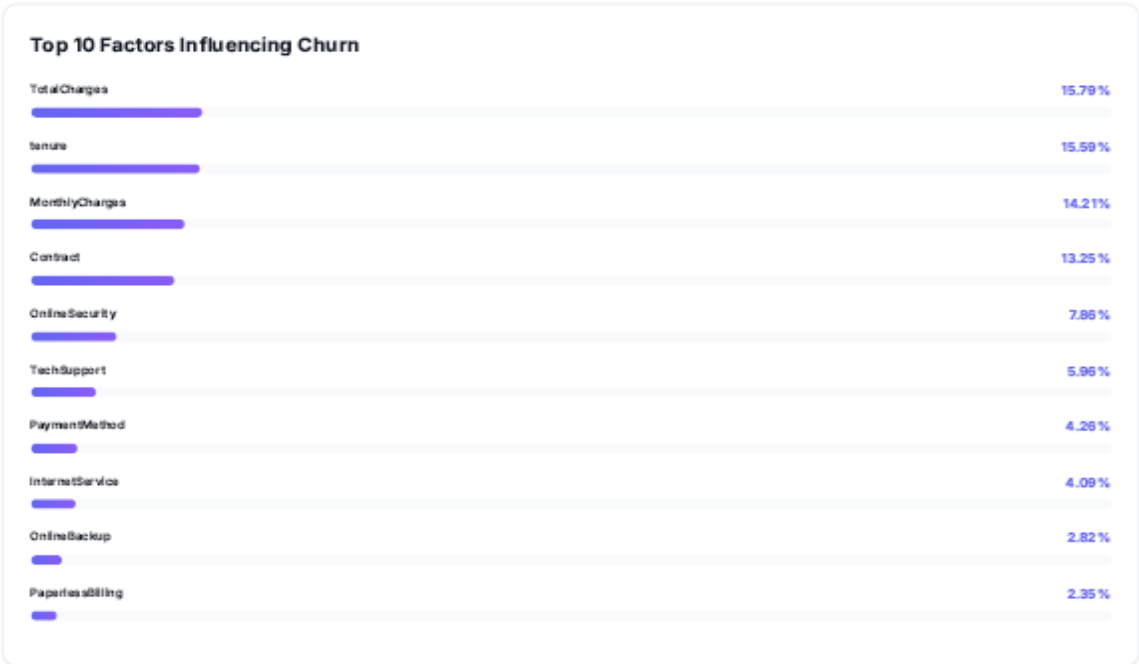
0488-GSLFR	1	\$69.50	Month-to-month	90.3%	Contact
0495-RVCBF	1	\$79.70	Month-to-month	90.2%	Contact
8149-RSOUN	1	\$93.85	Month-to-month	90.2%	Contact
7851-WZEKY	2	\$95.15	Month-to-month	90.1%	Contact
0107-YHINA	1	\$99.75	Month-to-month	90.0%	Contact
6230-BSUXY	1	\$85.00	Month-to-month	89.9%	Contact
4614-NUVZD	1	\$81.10	Month-to-month	89.9%	Contact
2004-OCQXK	1	\$81.95	Month-to-month	89.9%	Contact
4550-EVXNY	1	\$69.65	Month-to-month	89.9%	Contact
2636-ALXXZ	1	\$69.60	Month-to-month	89.9%	Contact
4871-JTKJF	1	\$69.65	Month-to-month	89.9%	Contact
8375-DKEBR	1	\$69.60	Month-to-month	89.9%	Contact
8779-QRDMV	1	\$39.65	Month-to-month	89.8%	Contact
8992-JQYUN	1	\$50.35	Month-to-month	89.7%	Contact
5178-LMXOP	1	\$95.10	Month-to-month	89.6%	Contact
7294-TMAOP	1	\$90.55	Month-to-month	89.6%	Contact
5797-APWZC	1	\$90.60	Month-to-month	89.6%	Contact
5134-IKDAY	1	\$69.80	Month-to-month	89.5%	Contact
1400-MMYXY	3	\$105.90	Month-to-month	89.5%	Contact
6023-YEBUP	3	\$100.95	Month-to-month	89.4%	Contact
9223-UCPVT	1	\$94.40	Month-to-month	89.3%	Contact
5167-ZFFMM	1	\$90.85	Month-to-month	89.2%	Contact
4695-WJZUE	2	\$88.55	Month-to-month	89.1%	Contact
3178-FESZO	1	\$100.25	Month-to-month	89.0%	Contact
3068-OMWZA	1	\$88.80	Month-to-month	88.9%	Contact

Customer churn risk analysis interface highlighting high-risk customers and factors that contribute to churn through analytical visualizations.



Key Insights & Feature Importance

Data-driven insights to guide retention strategies



Strategic Insights

Contract Type is Critical

HIGH IMPACT

Month-to-month contracts have 42.7% churn rate vs 6.8% for longer contracts.

Early Tenure Risk

HIGH IMPACT

Customers with <12 months tenure have 48.3% churn rate.

Tech Support Value

MEDIUM IMPACT

Customers without tech support churn at 41.6% vs 15.2% with support.

Payment Method Impact

MEDIUM IMPACT

Electronic check users have highest churn at 45.3%.

Recommended Actions

1

Implement Contract Incentive Program

Offer 15-20% discount for customers switching from month-to-month to annual contracts. Focus on customers with 6-12 months tenure.

2

Enhanced Onboarding for New Customers

Implement 90-day onboarding program with check-ins at 30, 60, and 90 days to increase early retention.

3

Tech Support Bundle Promotion

Offer free 3-month tech support trial to customers without support services, particularly those with high monthly charges.

4

Payment Method Migration Campaign

Incentivize electronic check users to switch to automatic payment methods with 5% discount or loyalty points.

Customer churn insights interface presenting important findings and summarized insights derived from the churn analysis.

4.5 Challenges Faced:

During the internship project, several practical challenges were encountered. Initial data preprocessing required careful handling of missing values, inconsistent data types, and categorical variable encoding to ensure accurate analysis. Converting features such as Total Charges into proper numeric format and managing imbalanced churn classes required additional validation and preprocessing steps.

Handling multiple customer attributes while identifying the most influential churn factors required systematic exploratory analysis and feature selection. Ensuring proper data splitting and avoiding overfitting during machine learning model training was also a key challenge. Model performance needed to be carefully evaluated using appropriate metrics rather than relying solely on accuracy.

Additionally, designing a clear and informative dashboard layout that effectively communicated business insights required multiple refinements. These challenges strengthened analytical thinking, improved technical problem-solving skills, and provided practical experience in handling real-world customer data for predictive analytics.

5 Learning Outcome

The internship significantly enhanced my technical and analytical competence in the domain of Data Science, Machine Learning, and Business Analytics. Through hands-on training, I developed proficiency in data preprocessing, feature engineering, exploratory data analysis, and predictive modeling techniques for customer churn analysis.

Working with real-world telecom datasets strengthened my understanding of data integrity, handling missing values, encoding categorical variables, and ensuring proper data validation. The integration of Python libraries such as Pandas, Matplotlib, and Scikit-learn for analysis and modeling, along with dashboard development tools for visualization, provided comprehensive exposure to practical data science workflows.

Furthermore, the experience deepened my understanding of classification algorithms, model evaluation metrics, and performance optimization techniques. Exposure to machine learning concepts within customer analytics broadened my perspective on how predictive modeling supports customer retention strategies and revenue optimization. Overall, the internship cultivated critical thinking, problem-solving ability, technical debugging skills, and the capacity to interpret data-driven insights for effective business decision-making.

6 Conclusion

The internship provided a valuable academic and technical experience, effectively bridging theoretical knowledge with real-world data science implementation. The Customer Churn Analytics project demonstrated the practical application of data analysis and machine learning techniques in understanding customer behavior and predicting churn patterns.

By combining data preprocessing, exploratory analysis, predictive modeling, and interactive dashboard visualization, the project reflected an integrated and structured analytical approach. The development of the churn prediction model and dashboard strengthened my technical foundation in Data Science and Machine Learning while enhancing my understanding of data-driven business intelligence.

This internship has equipped me with the practical skills and confidence required to undertake advanced analytics projects and contribute meaningfully to customer retention strategies, business optimization, and AI-driven decision-making processes in real-world industry settings.

7. Internship Completion Certificate



CERTIFICATE OF COMPLETION

DATE: 11-01-26

This is to Certify that Tarika A G has Successfully Completed a 15 days of Internship at Altruisty Innovation Pvt Ltd, from 16-12-25 to 31-12-25, in the Domain of Data science .

Throughout the Duration of the Internship, Tarika A G has Demonstrated Remarkable Growth and Development, Gaining Valuable Experience and Insights Into The Field of Data science .

Their commitment to learning and adapting to new challenges reflects Altruisty's core values of excellence and innovation.

We hereby acknowledge Tarika A G for This outstanding performance and dedication during the internship tenure.

Best Regards,

A handwritten signature in black ink, appearing to read "A. N. Bhavithra".

Bhavithra A N
HR & Executive Lead



REG:1252025583

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✉ altruistybusiness@gmail.com



Rajalakshmi Engineering College Thandalam, Chennai.

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Internship Feedback Form

Student Details:

Registration Number : _____
Student Name : _____
Department : Artificial Intelligence and Data Science
Year / Semester : _____
Internship Organization : _____
Trainer Name : _____
Internship Period : FROM : _____ TO: _____
Internship Duration : _____ DAYS

Section A: Student Feedback (to College)

1. Overall, how would you rate your internship experience?

☐ Excellent ☐ Very Good ☐ Good ☐ Satisfactory ☐ Poor

2. Did the internship help you gain useful knowledge or skills?

☐ Yes ☐ Partially ☐ No

3. What did you like most about your internship?

4. Remarks:

Student Signature: _____ Date: _____

Section B: Organization Feedback (to College)

1. Student's overall performance:

☐ Excellent ☐ Very Good ☐ Good ☐ Satisfactory ☐ Poor

2. Attitude, discipline, and professionalism:

☐ Excellent ☐ Very Good ☐ Good ☐ Satisfactory ☐ Poor

3. Contribution to assigned tasks/projects:

☐ Excellent ☐ Very Good ☐ Good ☐ Satisfactory ☐ Poor

4. Comments / Recommendations:

Organization Mentor/Manager Name & Designation: _____

Signature & Date: _____



**RAJALAKSHMI
ENGINEERING
COLLEGE**

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE
AND DATA SCIENCE**

Presented by

TARIKA A G

241801290

TITLE OF THE INTERNSHIP

REGISTRATION NUMBER : 241801290

STUDENT NAME : TARIKA A G

YEAR / SEMESTER : II/IV

NAME OF THE ORGANIZATION / COMPANY : ALTRUISTY INNOVATION PVT Ltd

ORGANIZATION ADDRESS : T.S. 107, Hindustan Teleprinters Staff Quarters,
SIDCO Industrial Estate, Guindy, Chennai, Tamil Nadu 600032

ONLINE / OFFLINE : ONLINE

TRAINER NAME : PAMAJULA PRAVALIKA

TRAINER'S CONTACT NO. :

INTERNSHIP PERIOD : FROM : 16/12/2025 TO: 31/12/2025

INTERNSHIP DURATION : 15 DAYS



About the Training

- Internship focused on Data Science and Machine Learning concepts.
- Hands-on training in data preprocessing, visualization, and clustering algorithms
- Developed Customer Churn Analytics Dashboard project using Python
- Built web application for real-time prediction using Flask
- Gained experience in real-world data analysis workflow
- Improved technical, analytical, and problem-solving skills



Abstract / Executive Summary

- This project focuses on developing a Customer Churn Analytics Dashboard to analyze customer data and identify patterns related to customer attrition. Data analysis and machine learning techniques were used to examine factors such as tenure, contract type, and monthly charges that influence customer churn. An interactive dashboard was created to visualize key metrics including churn rate, revenue loss, and customer segmentation. The system helps businesses understand churn behavior and supports data-driven decision-making to improve customer retention strategies.



Introduction

- **Customer Churn Analytics Dashboard** is a data-driven approach used to analyze customer behavior and identify customers who are likely to stop using a service. It helps businesses understand the factors influencing churn and take proactive measures to improve customer retention. In this project, data analysis and machine learning techniques were used to study customer data and identify patterns related to churn.
- **Key Points:**
 - Focus on analyzing customer churn patterns using data
 - Used Machine Learning techniques for churn prediction
 - Helps businesses reduce customer loss and improve retention strategies
 - Developed an interactive dashboard to visualize churn insights and key metrics



Objectives of the Internship

- > To gain practical knowledge in the **Data Science and Machine Learning domain**
- > To understand the **complete workflow of a real-world data science project**
- > To develop a **Customer Churn Analytics Dashboard** to analyze and monitor customer churn patterns
- > To learn **data preprocessing, exploratory data analysis, and data visualization techniques**
- > To implement **machine learning algorithms for customer churn prediction**
- > To gain hands-on experience with tools like **Python, data visualization libraries, and machine learning frameworks**
- > To improve **analytical thinking, programming skills, and problem-solving ability**



Brief Description of Work / Project

- **The project consists of the following modules:**
- **1. Data Processing**
 - Imported the telecom customer dataset containing attributes such as tenure, contract type, monthly charges, internet services, and payment methods.
 - Performed data cleaning and preprocessing to handle missing values and correct data types.
 - Selected relevant features related to customer behavior and churn analysis.
 - Prepared the dataset for further analysis and model development.
- **2. Model Development**
 - Implemented machine learning classification algorithms for customer churn prediction.
 - Split the dataset into training and testing sets for model evaluation.
 - Trained the model to identify patterns and factors influencing customer churn.
 - Generated churn predictions to identify customers with a high risk of leaving the service.
- **3. Data Visualization**
 - Created charts and graphs to visualize churn patterns and customer behavior.
 - Analyzed churn distribution based on contract type, tenure, and monthly charges.
 - Interpreted insights to understand key factors contributing to customer attrition.
- **4. Dashboard Development**
 - Developed an interactive Customer Churn Analytics Dashboard.
 - Integrated key metrics such as churn rate, revenue loss, and customer segmentation.
 - Designed a user-friendly interface to present insights through charts and visual reports.
 - Displayed churn analysis results to support data-driven business decisions.



Learning Outcome

- Gained practical knowledge in **Data Science and Machine Learning** concepts
- Learned **data preprocessing, exploratory data analysis, and data visualization** techniques
- Understood the **implementation of machine learning algorithms** for customer churn prediction
- Developed skills in **Python programming and machine learning libraries**
- Experienced building and presenting insights through an **interactive Customer Churn Analytics Dashboard**
- Improved **analytical thinking and problem-solving** abilities
- Understood **real-world applications of data-driven decision-making** in customer retention and business strategies



Conclusion

- The internship provided valuable hands-on experience in the field of Data Science and Machine Learning. The **Customer Churn Analytics Dashboard project** demonstrated how data analysis and machine learning techniques can be used to analyze customer data and identify patterns related to customer attrition.
- - Implemented machine learning techniques to analyze and predict customer churn
 - Identified key factors influencing churn to support business decision-making
 - Developed an interactive dashboard to visualize churn metrics and insights
 - Gained practical experience in data preprocessing, analysis, and predictive modeling
 - Demonstrated the real-world application of data science in improving customer retention and business strategies



INTERNSHIP CERTIFICATE



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Thank You

